

How Many of 780 Backtested Strategies Survive a Correction for Multiple Testing?

Z1: The Strategy Zoo

Pre-registration commit 01e84db | 44 ETFs | 780 configurations

1. Introduction

Anyone can produce a backtest with a good Sharpe ratio if they are allowed to try enough strategies and then report only the best one. This is a well documented problem in finance. *Harvey, Liu and Zhu* (2016) argue that a large part of the hundreds of factors published in the asset pricing literature would not survive a proper correction for how many factors were tested in the first place. *Bailey and Lopez de Prado* (2014) show the mechanism behind it: the expected maximum Sharpe ratio across N trials grows with N even when none of the strategies has any real skill.

My research question is as follows: **of a pre-registered set of 780 multi-asset strategy configurations, how many are still statistically significant once the number of trials is accounted for?**

I designed the project so that the answer would be checkable and not something I simply claim. The full list of 780 configurations was written down and committed to version control before a single backtest was run, at commit *01e84db*. This matters more than it might look. The entire calculation depends on the trial count being honest, so if configurations were added after seeing the results, or if unsuccessful ones were quietly dropped, every number in this paper would be meaningless.

Section 2 describes the data and the three tests. Section 3 gives the results, and Section 4 discusses what they mean and what I was not able to do.

2. Data and Method

The universe is 44 exchange traded funds across six buckets: equity, sectors, fixed income, commodities, real estate and FX. I pulled daily open, high, low, close and volume from yfinance, taking the longest history available for each ticker. SPY starts in 1993 and most of the others between 1996 and 2007. Before building anything on top of this data I cross-checked SPY, TLT and GLD against Stooq as a second source. Daily return correlations were 0.9986, 0.9978 and 0.99999. The SPY CAGR over 2000 to 2025 came out at 8.03%, which matches the well known published figure of 7 to 8%.

The grid has four signal families: time series momentum, cross-sectional momentum, short-term reversal, and momentum of realised volatility. Each family is crossed with a set of lookbacks, a skip-most-recent-month variant, five holding periods, three universe subsets and two sizing rules. That gives 780 configurations in total. Carry was deliberately left out, because a term structure carry signal needs a futures curve and I could not find a free source reliable enough to build one on.

Every configuration runs through the same engine: signal, target weight, turnover, costs, daily profit and loss. Transaction costs use the *Corwin and Schultz* (2012) estimator, which recovers a bid-ask spread from daily high and low prices alone. On its own it overestimates spreads badly for liquid names, giving roughly 27 bp for SPY where the true spread is under 1 bp, so I capped it with a liquidity ceiling based on each ticker's average dollar volume. The model is parameterised from published spread levels and not calibrated against real quote data, so its magnitudes are directional only. Results below are gross; for the best configuration net Sharpe is 0.66 against 0.67 gross, so costs change almost nothing here.

I then applied three tests to the full set of 780 trials:

Deflated Sharpe Ratio. Following *Bailey and Lopez de Prado* (2014), the best observed Sharpe is judged against the expected maximum under a null hypothesis of no skill, which depends on the trial count and the cross-sectional spread of trial Sharpes, rather than against zero. The test also corrects for the winning strategy's own skewness and kurtosis, which matters because momentum style returns are non-normal.

Multiple testing haircuts. Bonferroni, Holm, and Benjamini-Hochberg-Yekutieli thresholds applied to the p-values of all 780 trials. The last of these is the relevant one, since it controls the false discovery rate under arbitrary dependence, and these trials are heavily correlated with each other because of the same lookbacks and universes.

Reality Check. White's (2000) test, implemented with the stationary bootstrap of *Politis and Romano* (1994). Each trial has its own average return subtracted, which forces the null hypothesis of no skill onto the data. The returns are then resampled in random length blocks, with the same resampled date sequence applied to every trial at once, so that both the autocorrelation within a series and the correlation across trials survive the resampling.

3. Results

Across all 780 configurations the annualised gross Sharpe ratio has a mean of 0.20, a standard deviation of 0.19, a minimum of -0.46 and a maximum of 0.67. The best configuration is number 445: cross-sectional momentum, a 252 day lookback, semiannual rebalancing, the full 44 ticker universe and volatility targeted sizing. Taken on its own, a Sharpe of 0.67 out of a single backtest would look like a result worth showing. It turns out that it isn't.

Test	Result
Best observed Sharpe (annualised, gross)	0.67
Deflated Sharpe Ratio	0.639
Survivors, uncorrected at 5%	213 of 780
Survivors, Bonferroni	0
Survivors, Holm	0
Survivors, Benjamini-Hochberg-Yekutieli	0
White's Reality Check, p-value	0.190

213 of the 780 configurations clear an uncorrected 5% significance test, and none of them clear any of the three corrections. The Deflated Sharpe Ratio of 0.639 is far below the 0.95 that would be needed to call the best result real once the trial count and the non-normality of the returns are taken into account. The Reality Check agrees from a different direction: a p-value of 0.190 means that in roughly one bootstrap world in five, a set of 780 strategies with no skill at all produces a best result at least as good as the one I observed.

4. Discussion and Conclusion

The answer to my research question is zero. Nothing in this collection of 780 configurations survives a correction for how many configurations were tried. I predicted this before running the tests, because a maximum of 0.67 out of 780 attempts is not a large number, but the point of the project was to establish it properly rather than to guess it.

This is a useful outcome and not a failed one. The same 0.67 Sharpe, shown without the other 779 results next to it, would look like a modest but genuine strategy. The difference between the two presentations is not the backtest, which is identical in both. It is the audit trail: because the grid was committed before it was run, the trial count is verifiable by anyone who checks the repository history.

The work has real limitations. The universe is 44 liquid ETFs, which is the wrong place to look for several of the effects the literature documents most strongly in small and less covered stocks, so a null result here is not evidence that those effects do not exist anywhere. Carry is missing entirely for the data reasons given in Section 2. The Reality Check uses 500 bootstrap replications, enough for a p-value of 0.190 but too few if the result were near the threshold. All of the analysis is in-sample.

The last of those limitations is also the natural next step. My tests show that the best result is not unusual for a search of 780 tries, but they say nothing about a different question: if I had picked the winning configuration at the time and traded it forward, would it have kept working? Answering that needs combinatorially purged cross-validation and an estimate of the probability of backtest overfitting. That is left for further research.

References

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